

Comprehensive Landslide Inventory and Hotspot Analysis in the Chittagong Hill Tracts of Southeastern Bangladesh

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Abstract

The Chittagong Hill Tracts of southeastern Bangladesh, including the districts of Chittagong, Rangamati, Bandarban, Cox's Bazar, and Khagrachari, are acutely vulnerable to landslides due to their rugged topography, intense precipitation, and anthropogenic disturbances such as deforestation and unregulated hill cutting. To address the pressing need for accurate geospatial data in risk mitigation and urban planning, this study develops a comprehensive landslide inventory map encompassing 1,005 documented landslide occurrences. The dataset integrates inputs from NASA's global landslide inventory, peer-reviewed literature, field reports, and UAV drone surveys. Employing Getis-Ord Gi* within a GIS environment, the analysis delineates landslide hotspots and cold spots, thereby revealing critical spatial heterogeneity in landslide distribution. These results show the visual distribution of landslide inventory data and which area to focus on for the field survey to collect more landslide inventory points. These outputs are essential for enhancing disaster risk reduction strategies, informing land-use regulations, and fostering resilient infrastructure planning in this high-risk region of Bangladesh. While the study underscores the value of multi-source data aggregation, it also acknowledges inherent limitations such as discrepancies in data resolution and challenges associated with dynamic urban land use. Despite these constraints, the research contributes to the geohazard knowledge base. It serves as a pivotal resource for stakeholders involved in disaster risk reduction and sustainable land management in the CHT.

Keywords: Landslide inventory, Chittagong Hill Tracts, Getis-Ord Gi*, Hotspot analysis, GIS, Disaster management.

1 Introduction

Research on detailed landslide inventory and hotspot analysis has emerged as a critical area of inquiry due to the significant threat landslides cause to human life, infrastructure, and the environment worldwide. The Chittagong Hill Tracts in southeastern Bangladesh are highly at risk of landslides due to steep slopes, heavy monsoon rains and growing human activities. In recent years, tools like Remote sensing, GIS and Machine Learning have made it easy to map landslide risks locally, across entire regions and also globally (Jaedicke et al., 2013a; Wu et al., 2015). With more extreme weather and growing cities in very risky areas, accurate landslide mapping and hotspot detection have become more important day by day. In Chittagong Division, the monsoon rainfall increased by 8% due to climate change, which is causing 1 million people to be at greater risk (Ageenko et al., 2022). Deadly Landslides in 2007 and 2017, which killed 127 in Chittagong and 150 in Rangamati, show the urgent need for a better landslide susceptibility map for hazard planning and risk reduction (Chatterjee et al., 2024).

Landslide Research has advanced from basic models to powerful data-driven and hybrid methods combining physical science and machine learning (Strohmaier & Brenning, 2025). Today, Researchers use a mix of satellite data, UAVs, and field observations to build detailed inventories and improve risk mapping. New spatial tools like Getis-Ord Gi* help detect landslide hotspots and show how unevenly landslides occur across different areas (Jaedicke et al., 2013b) (Hölbling et al., 2016). Despite improvement, the development of universal and complete landslide inventories and tracking of hotspots through time continues to be a challenge. The majority of inventories suffer from such a problem with bad data quality, subject to limited coverage, and low time accuracy ("Landslide Inventory, Sampling and Effect of Sampling Strategies on Landslide Susceptibility/Hazard Modelling at a Glance," 2019). Data on previous monitoring results presents many times bias or gaps, particularly for slow-moving slides or in remote or inaccessible zones (Arbanas et al., 2014). The optimal approach in this sense to generate landslide inventory, data sample, and hot spots can still record much debate. There is an ongoing debate between researchers on whether automatic mapping or manual methods yield better results, and whether statistical or physically based models are more accurate for accurate analysis ((van Westen, n.d.);(Jaedicke et al., 2013b). Lack of reliable data and common analytical tools is still a

fundamental challenge, causing problems to predict and deal with landslide risks effectively. Developing countries face this challenge a lot, where this hazard is increasing with rapid urbanisation and climate change effects (Jaedicke et al., 2013b).

This analysis leads to three lines of thought (LOVs): generating comprehensive landslide inventories, studying spatial and temporal hotspots, and using multivariate data. Inventories provide critical information on the timing, date, and location of landslides, serving useful source for hazard and risk assessments (Jaedicke et al., 2013b); ("Statistical Modeling of Landslides," 2019). Areas of high landslide activity are detected by hotspot analysis, prioritizing risk-reduction efforts. (Hölbling et al., 2016). Combined use of diverse data sources enhances the accuracy and completeness of the inventories (Rabby & Li, 2019).

Many factors are responsible for landslides such as geology, slope, weather, heavy rainfall and human activities; they don't occur uniformly across all areas (Huang et al., 2022). Understanding these differences is key to managing risks, since landslide chances vary from place to place and can't be assessed with a one-size-fits-all method (Guzzetti, 2021)

In the Chittagong Hill Tracts, human activities have become major causes of landslides. Deforestation, hill cutting, unplanned housing and blocked drainage make slopes weaker and more prone to rain-triggered slides. The arrival of Rohingya refugees has added pressure by expanding settlements and using up natural resources in risky areas ("The Anthropogenic Aggravation of Landslide Disasters in Bangladesh," 2023). Considering all the risks, the whole thing created a complex situation that needs careful study and focused solutions.

2 Methodology

2.1 Study Area

The study area contains the Chittagong Hill Tracts (CHT) (Figure 1) of southeastern Bangladesh, and specifically covers the districts of Chittagong, Rangamati, Bandarban, Cox's Bazar, and Khagrachari. This region is about 13,295 square kilometres and represents the only part having mountains in Bangladesh, with elevations ranging from sea level to over 1,000 meters above mean sea level ("Chittagong Hill Tracts," 2025). The area has rugged topography with steep slopes (often more than 30°), and deep valleys formed by many rivers and streams (Hafsa et al., 2022).

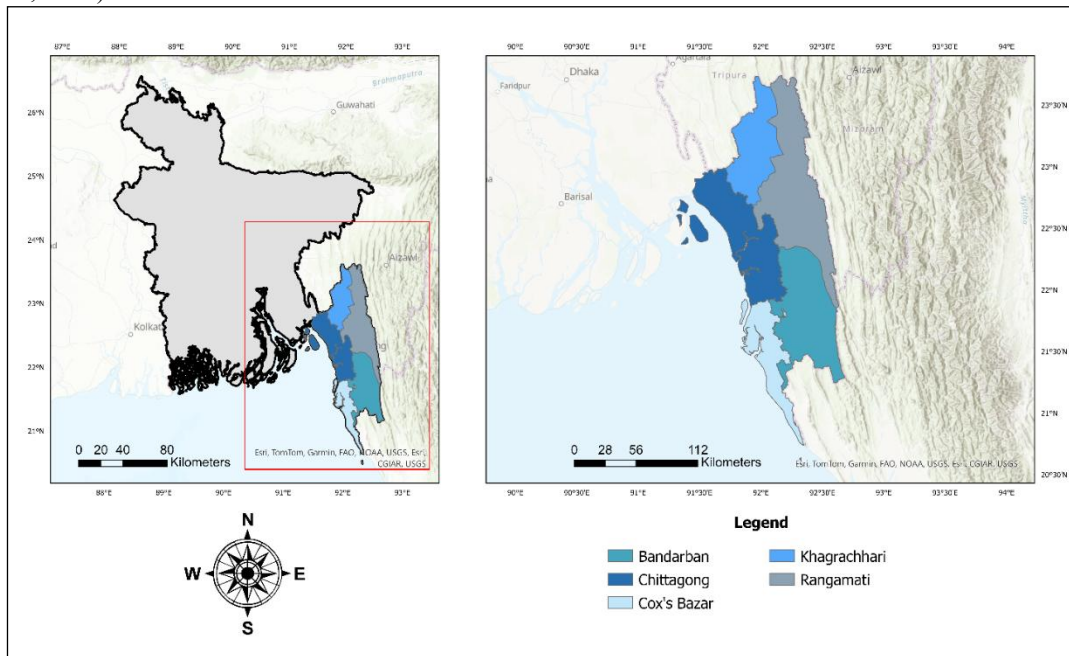


Figure 1 Study Area: The Chittagong Hill Tracts

The land in this area is mostly made up of soft rocks from the Tertiary period, such as sandstone, siltstone, and shale which are part of the Surma and Tipam groups. These rocks are usually loose or only slightly packed together, which makes them easily worn down by rain and wind, leading to unstable slopes (Jamshidi, 2023). The region has a tropical monsoon climate, with heavy yearly rainfall between 2500 and 4000 mm. More than 80% of the rain falls during the monsoon season, from June to October (Nadiruzzaman et al., 2025). The CHT region has undergone significant anthropogenic changes in recent decades, such as rapid population growth, unplanned urbanization, and massive deforestation. The population density has increased rapidly; in Chittagong city alone, 6558 families are now living in high-risk areas on 26 hills. Recent changes in land use have increased landslide vulnerability due to hill cutting, altering natural slopes, and removing vegetation (Chakma et al., 2023).

2.2 Data Collection

The landslide inventory data collection followed a multi-source approach to achieve broad spatial and temporal coverage. The integration approach used proven standards and practices for landslide inventory development, focusing on the quality, completeness, and reliability of the data (Höbling et al., 2016). NASA Global Landslide Catalogue (GLC) The NASA GLC offered standardized information on global landslide events (*LANDSLIDES @ NASA*, n.d.). The GLC, managed since 2007 at NASA Goddard Space Flight Centre, is the first and largest global compilation of landslide data (with more than 11,000 reports) and it is one of the key drivers of systematic landslide documentation. For the CHT area, the GLC dataset was retrieved from the Cooperative Open Online Landslide Repository (COOLR), which delivers georeferenced point entities of landslides along with their metadata. A systematic overview of peer-reviewed articles allowed us to gather and document landslide events, described with sound scientific care. The search was limited from 2000 to 2024, and the databases and search terms were: Web of Science, Scopus, and Google Scholar for keywords such as landslide in Bangladesh/Chittagong Hill Tracts/southeastern Bangladesh. We have identified 730 landslide inventory points from the study of (Rabby & Li, 2019). CEGIS has been involved in landslide assessment and mapping projects in the CHT region, contributing detailed inventory data with high spatial accuracy and local context (CEGIS, 2015). We have found some data from (CEGIS :: Home, n.d.). Official reports from local government agencies, disaster management organizations, newspapers, and media sources provided additional landslide occurrence data (*Landslide-Inventory-Report*, n.d.). High-resolution UAV surveys were conducted to capture landslide occurrence places. The surveys made use of DJI Mavic 2 Pro drones equipped with 20-megapixel cameras, used for a survey on the border road in Rajostholi in Figure 2.



Figure 2: UAV Drone Survey

2.3 Hot Spot Analysis

Hot spot and cold spot landslide areas were found through a spatial analysis, and the Getis-Ord G_i^* statistics were used for the analysis and implemented using ArcGIS Pro 3.0. The Getis-Ord G_i^* method was chosen as it is known to be effective in identifying spatial clustering patterns and has been successfully applied in a number of landslide studies for hotspot generation (Jaedicke et al., 2013b), (Höbling et al., 2016)

Type: Index (z-score) The Getis-Ord G_i^* statistic for a feature is calculated based on the local sum of a feature and its neighbours in comparison to the expected local sum of the feature if features were randomly distributed. The G_i^* statistic has the form, where,

$$G_i^* = \frac{\sum_{j=1}^n w_{i,j} x_j - \bar{X} \sum_{j=1}^n w_{i,j}}{S \sqrt{\frac{n \sum_{j=1}^n w_{i,j}^2 - \left(\sum_{j=1}^n w_{i,j}\right)^2}{n-1}}} \quad 1$$

Where: x_j is the attribute value for feature j , $w_{i,j}$ is the spatial weight between feature i and j , n is the total number of features. $\bar{X}$ is the mean of all attribute values, and S is the standard deviation of all attribute values. This statistic identifies statistically significant spatial clusters of high values (hot spots) and low values (cold spots) by evaluating each feature within the context of its neighbours.

3. Results

3.1 Inventory Map

The comprehensive landslide inventory map successfully documents 1,005 landslide occurrences across the Chittagong Hill Tracts region. This inventory represents a significant advancement in the documentation of landslide events for this vulnerable region, providing crucial spatial coverage that underpins susceptibility mapping and hazard assessments. The inventory compilation demonstrates the effectiveness of multi-source data integration, addressing the recognized need for extensive and multi-temporal landslide datasets. Figure 3 represents the robust Landslide inventory Map for the hilly region of Chittagong. The inventory map reveals the spatial distribution of landslide events across the five districts, with concentrations varying according to topographic, geological, and anthropogenic factors.

3.2 Hotspot Analysis Map

The Getis-Ord G_i^* analysis successfully delineated landslide hotspots and cold spots, revealing critical spatial heterogeneity in landslide distribution across the CHT. The hotspot analysis technique effectively captured landslide-prone areas at the regional scale, consistent with findings that show better performance of such techniques at national and regional scales compared to local scales.

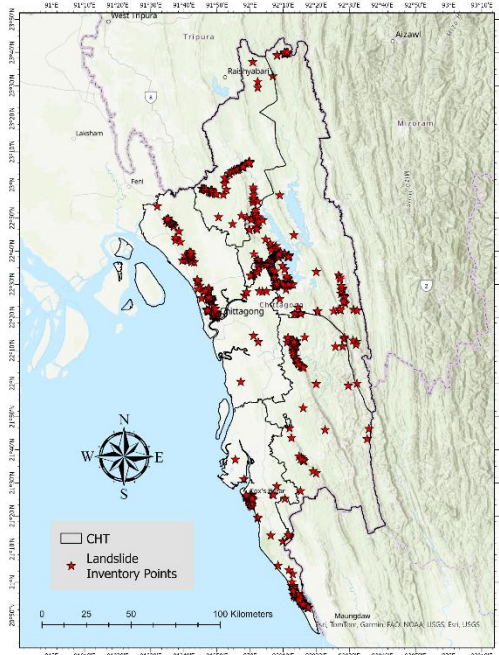


Figure 3: Landslide Inventory Map

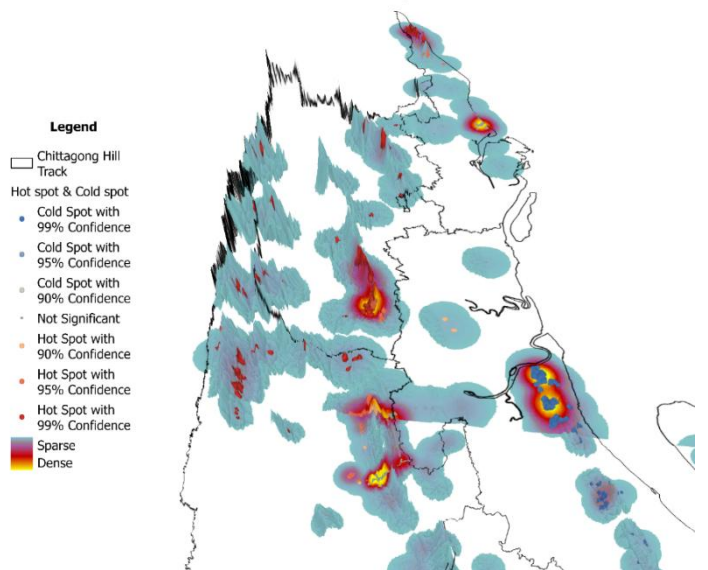


Figure 4: Hotspot and Coldspot

The analysis identified statistically significant clusters of landslide activity, providing valuable insights into the spatial patterns of landslide occurrence. These hotspots represent areas where landslide density is significantly

higher than would be expected by random distribution, indicating underlying controlling factors that promote landslide occurrence. The visualization of these patterns facilitates effective risk communication and informs targeted mitigation strategies. The spatial heterogeneity revealed through the analysis demonstrates the non-uniform distribution of landslide risk across the region. This finding is consistent with research showing that hotspot delineation effectively identifies spatial clusters of high landslide activity, though it often depends on the quality and resolution of input data.

4 Discussion

For the Chittagong Hill Tracts region, a detailed landslide inventory and hotspot analysis highlight a substantial methodological improvement in landslide hazard assessment. Combining multiple sources of data like NASA's global landslide inventory, peer-reviewed literature, first-hand field reports, newspapers, and UAV drone surveys forms a complete inventory for landslide ("Landslide Inventory, Sampling and Effect of Sampling Strategies on Landslide Susceptibility/Hazard Modelling at a Glance," 2019). The successful compilation of 1,005 landslide events indeed can be considered as a significant enhancement of the regional landslide database in terms of spatial and temporal scale compared to the previously available efforts in the area. The completeness of the inventory is especially important, as for many studies, event-based data and local datasets are used, resulting in variable completeness and little use for a regional-scale analysis. The multisource strategy used in this study increased the representativeness and the reliability of the assembled dataset, tackling important limitations in spatial coverage and in the temporal dimension. The spatial heterogeneity revealed through the Getis-Ord G_i^* analysis demonstrates the hotspots and coldspots of the historical landslide points (Jana & Sar, 2016). The concentration of hotspots in areas of intensive human activity, particularly around transportation corridors and urban centers, supports findings that adjacent to roads and slope undercutting are dominant anthropogenic triggers in tropical hill regions (Chatterjee et al., 2024). The comprehensive inventory and hotspot analysis provide a scientific foundation for evidence-based risk management in the CHT region, i.e., Landslide susceptibility mapping. The identification of statistically significant hotspot areas enables targeted resource allocation for monitoring, early warning systems, and structural mitigation measures. This landslide inventory map further facilitates the preparation process of the more accurate landslide susceptibility map in CHT. This integrated approach would support the development of more effective and equitable risk reduction strategies that address both physical and social dimensions of landslide vulnerability.

5 Conclusion

This study successfully developed a comprehensive landslide inventory encompassing 1,005 documented landslide occurrences in the Chittagong Hill Tracts of southeastern Bangladesh, representing a significant advancement in regional landslide documentation. The integration of multiple data sources, including NASA's global landslide inventory, peer-reviewed literature, field reports, and UAV drone surveys, effectively addresses the recognized challenges of incomplete or biased inventories in landslide research. The application of Getis-Ord G_i^* statistical analysis within a GIS environment successfully identified landslide hotspots and cold spots, revealing critical spatial heterogeneity in landslide distribution across the study region. This analysis provides valuable insights for targeted risk assessment and mitigation planning, supporting more informed decision-making for infrastructure development and disaster risk reduction. The research demonstrates the effectiveness of multi-source data integration approaches for landslide inventory compilation, addressing key limitations identified in the literature regarding data completeness and spatial-temporal coverage. The comprehensive nature of the inventory, combined with robust spatial analytical methods, provides a solid foundation for future landslide susceptibility mapping and hazard assessment in the region. The visual distribution of landslide inventory data and the identification of spatial hotspots will guide future field surveys and data collection efforts, facilitating the development of more robust landslide susceptibility maps. This research contributes to the broader goal of improving landslide risk assessment and management practices in vulnerable regions through enhanced data integration and analytical methods. Future research should focus on temporal analysis of landslide patterns, integration of additional conditioning factors, and validation of hotspot predictions through field verification. The methodology developed in this study provides a replicable framework for similar investigations in other landslide-prone regions, contributing to the advancement of landslide science and risk management practices.

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